

E004

Introduction to IT

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A marketing agency with 35 employees buys new laptops. The IT technician first installs the operating system, then a photo-editing program the designers use every day. The agency recently won a regional design award. Which of these is system software?

- A. The photo-editing program.
- B. The operating system.
- C. The award-winning design templates.
- D. The laptop's solid-state drive.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A dental office with three exam rooms has 12 PCs that must reach the internet provider's network. The office recently upgraded to 4K monitors. Which device forwards traffic between the office network and the provider's network?

- A. A switch.
- B. A wireless access point.
- C. A graphics card that supports the 4K monitors.
- D. A router.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Which device connects computers inside one local network and forwards frames to the correct port by MAC address?

- A. A switch.
- B. A router.
- C. A modem.
- D. A router, because all forwarding on any network is done by routers.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A regional distributor's database runs on a server in the company's basement and holds about 80,000 records. The Customers table uses CustomerID to identify each customer. The Orders table also has a CustomerID column showing who placed each order. In the Orders table, what is CustomerID?

- A. A foreign key.
- B. An index, because the table has 80,000 records.
- C. A composite key made of every column.
- D. The primary key of Orders.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A storm cut power to a clinic's server room, and staff could not open patient records for six hours. Logs show no records were viewed or changed during the outage. The clinic had passed its annual audit the month before. Which part of the CIA triad was affected?

- A. All three, because any incident breaks the whole triad.
- B. Confidentiality.
- C. Integrity.
- D. Availability.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A manager wants a control whose main purpose is to show whether a file has been changed since it was approved. Which control fits best?

- A. Adding a backup power supply to the server.
- B. Comparing the file's hash value to the approved hash.
- C. Encrypting the file at rest.
- D. Requiring a password, which guarantees no file is ever altered.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A warehouse team is building an inventory app. The project has a generous budget and a well-liked sponsor, and the manager wants developers to start coding Monday. Following the systems development life cycle, what should the team do first?

- A. Design the database and screens.
- B. Write the test plan and begin testing.
- C. Gather and analyze the users' requirements.
- D. Start coding Monday as the manager asked.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A startup founded by two former teachers plans to hire ten people next year. It wants to rent virtual servers from a cloud provider and choose and manage its own operating system and software, without buying any hardware. Which service model fits?

- A. Software as a Service (SaaS).
- B. Platform as a Service (PaaS).
- C. Infrastructure as a Service (IaaS).
- D. An on-premises data center.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A company uses a SaaS email platform. Who is mainly responsible for patching the email application itself?

- A. The SaaS provider.
- B. The customer patches the operating system; the provider patches the app.
- C. No one, because cloud software always updates itself.
- D. The customer's IT team.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

An online retailer's data team, which works remotely from three states, has 50,000 past emails, each tagged as spam or not spam. It wants a model that predicts whether new emails are spam. Which approach fits?

- A.** Unsupervised learning, because the data set is large.
- B.** Reinforcement learning.
- C.** Supervised learning.
- D.** Clustering the emails into natural groups.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A branch office has a 100 Mbps internet connection. It downloads a 500 MB software update that the vendor released last Tuesday. Ignoring overhead, what is the minimum download time?

- A.** 5 seconds
- B.** 40 seconds
- C.** 0.2 seconds
- D.** It cannot be estimated, because network speed always varies.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A 2 GB (2,000 MB) backup must finish transferring in 40 seconds. Ignoring overhead, what is the minimum connection speed needed?

- A.** 40 Mbps
- B.** 50 Mbps
- C.** 640,000 Mbps
- D.** 400 Mbps

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A marketing agency ~~with 35 employees~~ buys new laptops. The IT technician first installs the operating system, then a photo-editing program the designers use every day. ~~The agency recently won a regional design award.~~ Which of these is system software?

Target word: identify system software **Decides:** The operating system manages the hardware and runs other programs.

A	Confusion set	It performs a user task, which makes it application software.
B	Correct	Operating systems are the core of system software.
C	Camouflage	Templates are user files; the award detail does not decide anything.
D	Adjacent category	That is hardware, not software.

2. Answer D Concept 2

A dental office ~~with three exam rooms~~ has 12 PCs that must reach the internet provider's network. ~~The office recently upgraded to 4K monitors.~~ Which device forwards traffic between the office network and the provider's network?

Target word: device joining two networks **Decides:** Traffic crosses from one network to a different network.

A	Confusion set	A switch connects devices inside the same local network.
B	Adjacent category	It lets wireless devices join the local network; it does not route between networks.
C	Camouflage	Answers the monitor detail, which has nothing to do with networking.
D	Correct	Routers forward packets between separate networks.

3. Answer A Concept 2 · transfer

Which device connects computers inside one local network and forwards frames to the correct port by MAC address?

Target word: device within a single network — reverse of the router item **Decides:** Forwarding by MAC address inside one LAN.

A	Correct	Switches forward frames among devices on the same LAN by MAC address.
B	Inverted operation	Routers join different networks and forward by IP address.
C	Adjacent category	A modem converts signals for the provider's line; it does not switch between local devices.
D	Absolute language	Switches handle forwarding inside a local network.

4. Answer A Concept 3

A regional distributor's database ~~runs on a server in the company's basement and holds about 80,000 records.~~ The Customers table uses CustomerID to identify each customer. The Orders table also has a CustomerID column showing who placed each order. In the Orders table, what is CustomerID?

Target word: role of CustomerID in Orders **Decides:** It points to another table's primary key.

A	Correct	It links each order to a row in the Customers table.
B	Camouflage	Table size does not decide the column's role.
C	Adjacent category	A composite key combines columns to identify a row; that is not what CustomerID does here.
D	Confusion set	A customer can place many orders, so CustomerID cannot uniquely identify an order.

5. Answer D Concept 4

A storm cut power to a clinic's server room, and staff could not open patient records for six hours. Logs show no records were viewed or changed during the outage. ~~The clinic had passed its annual audit the month before.~~ Which part of the CIA triad was affected?

Target word: which CIA principle **Decides:** Staff lost access; nothing was seen or altered.

A	Absolute language	Incidents can affect one principle without touching the others.
B	Confusion set	No one saw data they should not have.
C	Adjacent category	The logs show no data was changed.
D	Correct	The records existed and were intact but could not be reached.

6. Answer B Concept 4 · transfer

A manager wants a control whose main purpose is to show whether a file has been changed since it was approved. Which control fits best?

Target word: control that protects integrity **Decides:** Detecting change is integrity.

A	Confusion set	That protects availability.
B	Correct	A changed file produces a different hash, revealing tampering or corruption.
C	Adjacent category	Encryption mainly protects confidentiality.
D	Absolute language	Authorized users can still change files; passwords do not guarantee integrity.

7. Answer C Concept 5

A warehouse team is building an inventory app. ~~The project has a generous budget and a well-liked sponsor,~~ and the manager wants developers to start coding Monday. Following the systems development life cycle, what should the team do first?

Target word: first SDLC activity **Decides:** Requirements must be known before design or build.

A	Dropped element	Skips requirements; design needs them as input.
B	Inverted operation	Testing comes after building.
C	Correct	Planning and analysis define what the system must do.
D	Camouflage	The manager's eagerness does not change the phase order.

8. Answer C Concept 6

A startup ~~founded by two former teachers plans to hire ten people next year.~~ It wants to rent virtual servers from a cloud provider and choose and manage its own operating system and software, without buying any hardware. Which service model fits?

Target word: cloud model by control layer **Decides:** The customer manages the OS on rented infrastructure.

A	Confusion set	SaaS delivers a finished app; the customer controls almost nothing below it.
B	Adjacent category	PaaS hides the OS; the customer manages only code and data.
C	Correct	IaaS rents compute, storage, and networking; the customer manages the OS and up.
D	Inverted operation	That means buying hardware, which the startup wants to avoid.

9. Answer A Concept 6 · transfer

A company uses a SaaS email platform. Who is mainly responsible for patching the email application itself?

Target word: responsibility under SaaS — reverse direction **Decides:** In SaaS the provider runs the application.

A	Correct	The provider manages the application, platform, and infrastructure.
B	Confusion set	Describes a split closer to IaaS; SaaS customers do not manage the OS.
C	Absolute language	Someone is responsible; in SaaS it is the provider.
D	Inverted operation	That would be true for software the customer installs itself.

10. Answer C Concept 7

An online retailer's data team, ~~which works remotely from three states,~~ has 50,000 past emails, each tagged as spam or not spam. It wants a model that predicts whether new emails are spam. Which approach fits?

Target word: learning type **Decides:** The training data is labeled with the answer.

A	Camouflage	Size does not decide the method; the labels do.
B	Adjacent category	That learns from rewards through trial and error, not from labeled examples.
C	Correct	Labeled examples train a model to predict a known outcome.
D	Confusion set	Clustering is unsupervised and ignores the labels the team already has.

11. Answer B Concept 8

A branch office has a 100 Mbps internet connection. It downloads a 500 MB software update ~~that the vendor released last Tuesday.~~ Ignoring overhead, what is the minimum download time?

Target word: transfer time **Decides:** $500 \text{ MB} \times 8 = 4,000 \text{ megabits}$.

A	Dropped element	Divides 500 by 100 and skips the 8 bits per byte.
B	Correct	$4,000 \text{ Mb} / 100 \text{ Mbps} = 40 \text{ s}$.
C	Inverted operation	Divides speed by size.
D	Absolute language	A minimum time can be calculated from the rated speed.

12. Answer D Concept 8 · transfer

A 2 GB (2,000 MB) backup must finish transferring in 40 seconds. Ignoring overhead, what is the minimum connection speed needed?

Target word: solve for speed **Decides:** $2,000 \text{ MB} \times 8 = 16,000 \text{ megabits}$.

A	Wrong stem number	The time limit from the stem, not a speed.
B	Dropped element	$2,000 / 40$ forgets to convert bytes to bits.
C	Inverted operation	Multiplies by the time instead of dividing.
D	Correct	$16,000 \text{ Mb} / 40 \text{ s} = 400 \text{ Mbps}$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	System vs. application software	System software (the operating system, drivers, utilities) runs the hardware; application software does a user's task on top of it.	Calling any program that came with the computer system software.	"Manages hardware" or "runs other programs" vs. "lets the user do a task."
2	Router vs. switch	A switch connects devices within one local network; a router forwards traffic between different networks.	Treating the two as interchangeable because both "connect things."	"Between the office network and the internet" vs. "among the office PCs."
3	Primary and foreign keys	A primary key uniquely identifies each row in its own table; a foreign key is a column that points to another table's primary key.	Calling any ID column a primary key regardless of which table it sits in.	The same ID column appearing in two tables.
4	CIA triad	Confidentiality = only authorized people see data; integrity = data is not altered improperly; availability = data and systems are reachable when needed.	Labeling every outage or incident a confidentiality breach.	What was lost: secrecy, accuracy, or access.
5	SDLC phase order	Planning and requirements analysis come before design, which comes before building, testing, and deployment.	Starting to code before requirements are known.	"What should the team do first?"
6	Cloud service models	IaaS rents infrastructure (customer manages OS and apps); PaaS rents a platform (customer manages code and data); SaaS rents a finished application (provider manages nearly everything).	Thinking the customer manages the same layers in every model.	Which layer the customer wants to control.
7	Supervised vs. unsupervised learning	Supervised learning trains on labeled examples to predict a known outcome; unsupervised learning finds patterns in unlabeled data.	Choosing the method by data size instead of whether labels exist.	Data that is "tagged" or "labeled" with the answer.
8	Bits vs. bytes	Network speed is in bits per second and file size in bytes; multiply bytes by 8 before dividing by speed.	Dividing megabytes by megabits per second directly.	"Mbps" next to "MB" or "GB."

Course scope (public description): Introduction to IT introduces the fundamental concepts, structures, and roles of information technology (IT) within an organization. The course is structured to provide a comprehensive understanding of the various aspects of IT, including networking, database management, cybersecurity, artificial intelligence (AI), and software development. This course also explores the essential components of the typical technology infrastructure for various types of organizations, the role and interaction of each component, and the evolving nature of technology and its implications for business practices. There are no prerequisites for this course.